

SHIVAM CHAUDHARY

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RESEARCH INTERESTS

How sensorimotor cortex handles the unexpected. I study volitional control of locomotion in freely moving macaques — what motor and somatosensory cortex do when a walking animal has to deal with a perturbation it did not see coming. Most of my work sits at the point where the neuroscience meets the hardware: real-time control systems, closed-loop behavioral rigs, chronic electrophysiology, and the decoding pipelines that make the loop close fast enough to matter.

EDUCATION

PhD Biomedical Engineering, Brown University, RI, USA (2026 - Present)

Advisors: Prof. David Borton and Prof. Wilson Truccolo

Dissertation: Volitional Control of Locomotion in Non-Human Primates (working title)

MSc Cognitive Science, Indian Institute of Technology (IIT) Gandhinagar, India (2021 - 2023)

Advisor: Prof. Krishna Prasad Miyapuram, Brain and Informatics Lab

Dissertation: [EEG-based Brain-Computer Interface for Grasp Motion Prediction](#)

Grade: 8.97 / 10.00 (US equivalent: 3.81 / 4.00). Earned an A+ (11/10) in Computational Neuroscience.

BSc Information Technology, St. Xavier's College (Autonomous), Mumbai, India (2017 - 2020)

Coursework: C, C++, Python, Java, Machine Learning, Deep Learning, Network Security, Web and Mobile Dev

Grade: 3.61 / 4.00

RESEARCH EXPERIENCE

Research Associate, Borton Lab & Truccolo Lab, Brown University (Nov 2025 - Aug 2026)

Built out the hardware and software backbone for a study on how primate sensorimotor cortex adapts to perturbations during walking, with Prof. David Borton and Prof. Wilson Truccolo.

Designed a closed-loop behavioral rig that reads gait phase off live pose estimates and acts on it in real time — dispensing reward on stride initiation and moving a motorized obstacle during swing phase. Runs on a DeepLabCut-Live pipeline at roughly 10-20 ms end-to-end.

Stood up a six-camera GigE Vision kinematics system (mvBlueCOUGAR-XD, 10 GbE, hardware-triggered at 100 Hz). Chased down frame drops to packet delay, NIC ring buffer behavior, and transmission stagger; now writing a Python/NVENC capture pipeline (Harvester + GenTL, CUDA debayer, hardware HEVC encode) to replace the vendor software.

Set up dual-NSP Blackrock recording with CerePlex W wireless headstages, synced the two units on a shared TTL pulse, and wired peripheral hardware — pellet dispenser, stepper-driven gantry, limit switches — into the neural data stream so events land on the same clock as the spikes.

Ported experiment control to QNX RTOS: brought up QNX SDP 8.0 on ARM, wrote a GPIO resource manager in C, and validated an Intan RHS2116 EMG breakout over SPI.

Handled the animal side too — marker tattooing for markerless and marker-based kinematics, IACUC-compliant consumables, and day-to-day rig maintenance.

Research and Development Engineer I, University of California, Berkeley (Sep 2023 - Nov 2025)

Worked with Prof. Preeya Khanna and Dr. Woohyun Kim on how the brain runs sensorimotor control across sequences of dexterous movements.

Built a feedback-dependent dexterous manipulation task from scratch — hardware, Python control software, and experiment design — used for both human behavioral work and primate electrophysiology.

Wrote the neural data analysis pipelines and ran a coherence analysis between dorsal premotor and primary somatosensory cortex to get at the dynamics of dexterous control.

Trained naive rhesus macaques from the ground up, assisted in surgical procedures, and shared animal care duties.

Graduate Researcher, Brain and Informatics Lab, IIT Gandhinagar (Aug 2021 - Aug 2023)

Master's dissertation. Built an EEG-based BCI that classifies hand open versus close. Wrote the spectral feature extraction and classification code in Python, hit above-chance within-subject accuracy, and put together a pipeline for real-time control of a virtual and 3D-printed prosthetic hand. (Prof. Krishna Prasad Miyapuram)

Drum beats and brain rhythms (May 2022 - Jul 2022)

Looked at how high-density EEG tracks drumbeats, with Prof. Derek Lomas and Prof. Miyapuram. Designed the experiment, collected the data, and found participant-specific neural signatures for individual music pieces. Presented at CODS-COMAD '23.

Cable-driven upper-limb exoskeleton (Aug 2022 - Nov 2022)

Designed, built, and wrote control code for a wearable cable-driven exoskeleton aimed at upper-limb assistance, with Prof. Vineet Vashishta.

PUBLICATIONS

Peer-Reviewed Papers

S. Chaudhary, P. Pandey, K. P. Miyapuram, and D. Lomas, "Classifying EEG signals of mind-wandering across different styles of meditation," *Brain Informatics*, vol. 4, no. 1, pp. 152-163, Mar. 2022, doi: [10.1007/978-3-031-15037-1_13](https://doi.org/10.1007/978-3-031-15037-1_13). Presented at University of Padova, Italy, July 2022 — [PDF](#)

P. Pandey, P. Gupta, **S. Chaudhary**, K. P. Miyapuram, and D. Lomas, "Real-time sensing and neurofeedback for practicing meditation using simultaneous EEG and eye tracking," *2022 IEEE Region 10 Symposium (TENSYP)*, 2022, pp. 1-6, doi: [10.1109/TENSYP54529.2022.9864414](https://doi.org/10.1109/TENSYP54529.2022.9864414). Poster at IIT Bombay, July 2022 — [PDF](#)

S. Singh, P. Pandey, **S. Chaudhary**, K. P. Miyapuram, and J. Lomas, "Towards the development of personalized and generalized interfaces for brain signals across different styles of meditation," *Proc. 13th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, Gandhinagar, India, 2022, pp. 54-62, doi: [10.1145/3571600.3571656](https://doi.org/10.1145/3571600.3571656) — [PDF](#)

Extended Abstracts and Short Papers

S. Soni, **S. Chaudhary**, and K. P. Miyapuram, "Enhancing motor imagery based brain-computer interfaces for stroke rehabilitation," *Proc. 7th Joint International Conference on Data Science & Management of Data (11th ACM IKDD CODS and 29th COMAD)*, 2024.

S. Chaudhary, K. P. Miyapuram, and D. Lomas, "Predicting drum beats from high-density brain rhythms," *Proc. 6th Joint International Conference on Data Science & Management of Data (CODS-COMAD '23)*, ACM, New York, NY, USA, 2023, pp. 291-292, doi: [10.1145/3570991.3571029](https://doi.org/10.1145/3570991.3571029) — [PDF](#)

V. Rohira, **S. Chaudhary**, S. Das, and K. P. Miyapuram, "Automatic epilepsy detection from EEG signals," *Proc. 6th Joint International Conference on Data Science & Management of Data (CODS-COMAD '23)*, ACM, New York, NY, USA, 2023, pp. 272-273, doi: [10.1145/3570991.3570995](https://doi.org/10.1145/3570991.3570995) — [PDF](#)

CONFERENCE PRESENTATIONS

Poster, Society for Neuroscience (SfN '24), Chicago, 2024 [[PSTR287.20](#)]

W. Kim, **S. Chaudhary**, T. Oppenheim, and P. Khanna; "A paradigm to study the role of tactile contact events in learning and execution of object manipulation behavior."

HONORS AND AWARDS

International Travel Grant, IIT Gandhinagar (2022)

Rs. 1,20,000 (approx. USD 1,400) to present at the University of Padova, Italy.

TEACHING AND MENTORING

Teaching

Lead Teaching Assistant, Computational Neuroscience, Neuromatch Academy (Jul 2023)

Ran a team of TAs through an advanced computational neuroscience course and helped shape the course material.

Teaching Assistant, Computation and Cognition, IIT Gandhinagar (Aug 2022 - Nov 2022)
Walked students through programming and problem-solving from a cognitive science angle. Gave a lecture on analyzing event-related potentials.

Teaching Assistant, Computational Neuroscience, IIT Gandhinagar (Jan 2022 - May 2022)
Advised student projects on music, meditation, and BCIs; graded assignments; taught a session on EEG signal processing and classification with a live demo.

Mentoring (all IIT Gandhinagar unless noted)

Saher Soni (MSc) (Jan 2023 - May 2023)
Decoding motor imagery from EEG with LSTMs. Manuscript accepted at CODS-COMAD 2024. *Now a PhD student at IIT Gandhinagar.*

Smriti Saini (MSc) (Jan 2023 - May 2023)
Classifying hand postures (reaching, twisting, grasping) from surface EMG. *Now a PhD student at MIT Brain and Cognitive Sciences.*

Dheemant Jallepalli (Undergraduate, IIT Jodhpur) (Jan 2023 - Jul 2023)
Separating EEG signals for winning versus losing a gamble after a decision. *Now a Research Assistant at Harvard University.*

Riddhi Johri (MSc) (Jan 2023 - May 2023)
Classifying imagined speech from EEG using machine learning and deep learning. *Now a Software Engineer at NXP.*

Bagmish Sabhapondit (MSc) (Jan 2023 - May 2023)
Predicting drum beats from high-density brain rhythms and stimulus-response correlations in EEG.

INVITED TALKS

EEG Signal Processing and Classification, IIT Gandhinagar (Mar 2023)
Invited by Prof. Krishna Prasad Miyapuram; talk plus a live demo.

Event-Related Potentials in EEG, IIT Gandhinagar (Nov 2022)

Introduction to Brain-Computer Interfaces, EETI Foundation (Nov 2022)
Online talk to roughly 100 participants.

ACADEMIC SERVICE

Reviewer, IEEE DSAA (2026)

Reviewer, Cognitive Science Society (2024, 2025)

Judge, Neurotech @ Berkeley Hackathon, UC Berkeley (2025)

Reviewer, ICONIP (2022)

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, PyTorch/TensorFlow, pyserial), C, C++, MATLAB, React (web and native), LaTeX
Neural Recording	Blackrock NSP / CereLink / CerePlex W, Intan RHS2116, EGI 64-channel EEG, TMSi EMG, Lab Streaming Layer
Real-Time Systems	QNX RTOS (SDP 8.0), closed-loop control, GenICam/GigE Vision, DeepLabCut-Live, CUDA/NVENC pipelines, hardware triggering and TTL sync
Behavior & Kinematics	VICON and SIMI motion capture, multi-camera markerless tracking, PsychoPy, operant task design
Hardware	PCB design, mechatronics, stepper/servo control, optoisolated I/O, CNC milling, 3D design (Fusion 360), 3D printing
Animal Research	Rhesus macaque positive-reinforcement training, surgical assistance, IACUC protocol compliance, animal care
Other	GCP, AWS, MongoDB, Git

PROFESSIONAL DEVELOPMENT

Brain, Computation, and Learning (BCL) Workshop, Indian Institute of Science, Bengaluru (Jan 2023)
Sessions with faculty working on invasive BCI, including Prof. Rajesh P. N. Rao.

Measurement and Analysis of Human Locomotion, 5-day course with Dr. Kamiar Aminian, EPFL

Computational Neuroscience, Neuromatch Academy (2021)

BCI & NeuroTechnology Spring School, g.tec (2021)

Certifications: Building Transformer-based NLP Applications, NVIDIA DLI ([cert](#)); TensorFlow Developer Professional and Deep Learning Specialization, deeplearning.ai ([cert](#))

INDUSTRY EXPERIENCE

Full Stack Developer, Prodio Designworks, Mumbai (Apr 2020 - Sep 2020)
MERN-stack developer on an agile team. Main project was an internal web app for the International Congress of Oral Implantologists (ICOI) — ReactJS/Redux front end, NodeJS back end — covering member management, payment scheduling, and credentialing.

Bachelor's Dissertation, St. Xavier's College (Autonomous), Mumbai (Oct 2019 - Apr 2020)
Worked in a pair on classifying human emotions from EEG under Prof. Lydia Fernandes, using wavelet transforms, PCA, and SVMs to get well above chance.

LEADERSHIP AND OUTREACH

Website Lead, G20-Ignite Sci-Tech Fair, IIT Gandhinagar (Mar 2023 - Apr 2023)
Led a team building the site for the G20-sponsored [Ignite](#) fair, aimed at school children.

Website and Application Manager, Technical Council, IIT Gandhinagar (May 2022 - Jul 2023)
Led a team modernizing the student body's technology infrastructure, including a React Native festival app now on the Play Store.

Volunteer Teacher, Door Step School (NGO), Mumbai (Dec 2017 - Jun 2018)
Taught social science, English, and life skills to students from underserved families.

REFERENCES

[Prof. David Borton](#)

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Brown University
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[Prof. Wilson Truccolo](#)

Associate Professor of Computational Neuroscience
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[Prof. Preeya Khanna](#)

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